

THE SIMULATION LOOP

carried out before any data exist

1 Build a generative model

Write the hypothesis as a formula. Set the numbers.

Which processes generate y , and how large are they?

2 Simulate and visualise

Generate and plot the data the model implies.

Could my system really produce these data?

3 Iterate the design

Replay the study. Vary n , predictor range, effort.

Is the interval narrow enough for my claim?

revise and repeat

a design you can defend

Research question

Hypothesis

Design and collect data

Analyse

Conclude

new questions

① wrong estimand

② post hoc decisions

③ untestable hypothesis

④ detect, not estimate

⑤ uninformative sampling

⑥ error and dependence

⑦ model mismatch

⑧ threshold conclusions

⑨ significance filter

● pitfall a design choice can fix

○ pitfall the loop exposes but cannot resolve

numbered as in Box 1